Python Basics & Data Types

1. What is Python?

Python is an interpreted, high-level and general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability with its notable use of significant indentation.

Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

2. Key Features of Python

- **Easy to Learn and Use:** Python has a simple syntax similar to the English language.
- **Interpreted Language:** Python code is executed line by line, which makes debugging easier.
- **Dynamically Typed:** You don't need to declare the type of a variable when you create one.
- **Extensive Standard Library:** Python comes with a large standard library that supports many common programming tasks.

3. Variables and Data Types

Variables are containers for storing data values. In Python, variables are created when you assign a value to it.

Built-in Data Types:

- **Text Type:** str
- **Numeric Types:** int, float, complex
- **Sequence Types:** list, tuple, range
- **Mapping Type:** dict
- **Set Types:** set, frozenset
- **Boolean Type:** bool

4. Input/Output and Type Conversion

Output: The print() function prints the specified message to the screen.

Input: The input() function allows user input. By default, it returns a string.

Type Conversion (Casting): You can convert from one type to another with the int(), float(), and str() methods.

5. Operators in Python

Operators are used to perform operations on variables and values.

- **Arithmetic:** +, -, *, /, // (floor division), % (modulus), ** (exponentiation)
- **Comparison:** ==, !=, >, <, >=, <=
- **Logical:** and, or, not